



Adoption of modern digital innovations: a systematic review of social impacts in online community forums

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Abstract

Scholars have employed multiple methodologies to examine the correlation between different social networks and the extent of technological adoption among consumers. Although there has been a lack of uniformity in understanding and operationalising fundamental networking site constituents, the adoption and implementation of said analytical paradigms have yielded a wealth of revelations. To address these issues, this research examines the relationship between social networks and consumer adoption of digital technologies, addressing inconsistencies in defining and operationalising key components of social media platforms. Using the Community, Interference, Correlation, and Results (CICR) framework, the study evaluates existing paradigms and their limitations, including inconsistencies in cognitive models and research scenarios. Key recommendations include developing a systematic methodology for investigating the correlation between social media connectivity and technology adoption, considering socio-demographic variables as moderators and behavioral attributes as mediators. The study examines the CICR model's limitations, including inconsistencies in cognitive paradigms, research scenarios, survey focus areas, and recent advancements. It proposes a systematic methodology to explore the correlation between social media connectivity and consumer technology adoption within the information communication technology (ICT) context, covering hardware, software, and devices such as smartphones, computers, and related applications. This research emphasizes the role of social media in shaping consumer attitudes toward digitalisation and highlights its implications for advancing ICT.

Keywords Competitive analysis · Consumer technologies · Online communities · Observational studies of literature · Social life

1 Introduction

According to a growing corpus of studies, modern technology adoption is a critical determinant of effective data systems (Ortega et al. 2022; Chew et al. 2022; Tsai 2017). As a result of this realisation, scholars have focused their efforts on identifying the elements that impact people's acceptance of consumer technology (Abouzahra and Ghasemaghahi 2022). Researchers in both the field of communication management and the field of behavioural science have taken an interest in consumers' use of advanced technologies since it is frequently regarded as an expression of behaviour. The authors of a study (Pirani et al. 2023) contend that an individual's point of view on a certain conduct is an essential component in describing that behaviour. To paraphrase what was said by Hoffmann and Akbar (2023), authors believe that both internal and outside factors have an effect on the perceptions that clients have of a company. The assumption that consumers' perceptions of innovation mirror their

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individual levels of use is supported by a growing amount of scholarly data (Mc Hugh et al. 2023; De Vos et al. 2022; Baierl et al. 2022).

The technological acceptability paradigm (TAP) (Oyman et al. 2022; Katebi et al. 2022), is the dominant frame of reference that is commonly used to forecast the use of client technologies. For the purpose of measuring consumer perception, TAP employs two fundamental indicators, namely perceived usability (PU) and perceived ease of usage (PEOU). The degree to which a person perceives that making use of a platform would improve productivity is what is meant by the term PU. On the other hand, the degree to which a person feels that making use of the platform will be comparatively effortless is what is meant by the term PEOU. The simplicity of TAP's description of why some people embrace new technologies has contributed to its widespread adoption. Secondly, TAP's focus on data processes makes it a viable framework for explaining how IT is used by people of all backgrounds and in all parts of the world, using a broad range of technologies. Finally, there is substantial empirical validation for the aggregate pblackictive performance of TAP, that is supported by a solid conceptual foundation that includes thoroughly investigated and established psychometric indicators (Natasia et al. 2022; Usman et al. 2022).

TAP has been criticised for not being able to take into account enabling conditions as a factor in tech adoption (Ly et al. 2023; Thomas et al. 2023; Malatji et al. 2020), but this doesn't seem to take away from the model's success in general. The argument put forth is that community values could be viewed as independent parameters since they are embedded in the results (Lin et al. 2020; Rezaei et al. 2019). Researchers have mostly employed TAP in workplaces where staff members have to utilise a new technological device before they are allowed to opt out of doing so Shin et al. (2019); Ninaus et al. (2021). Furthermore, TAM is primarily conceived as a scheme to clarify personal decisions, although technological choices are frequently made in concert with others (Alshurideh et al. 2023). Hence, TAP's failure to account for cultural factors is a major flaw. Researchers have proposed that an individual's behaviour is most accurately characterised by their involvement with community organisations rather than their absence from them Putranto et al. (2023); Tharkude (2023).

Academics have proposed that a person makes decisions as a member of a community unit, and that connections between members of that community can lead to increased consumer participation and devotion (Santos et al. 2022). Furthermore, social media platforms encourage communication and collaboration between members of the group (Abbas et al. 2022). Professional productivity (Zimmer 2022), staff turnover (Boland et al. 2023), creativity (Zimmer 2022), immoral practises (Khan et al. 2023),

entrepreneurs (Lacarcel and Huete 2023), and consumer gadget use are only some of the behavioural effects that have been explained by research into the impact of social relationships via social media platforms (Lim and Rasul 2022; Ahmed et al. 2022).

Information communication technologies (ICT) serves as a broad term encompassing a diverse array of technologies that empower users to generate, access, distribute, store, manage, and exchange information in a digital framework. This encompasses computer hardware, software applications, mobile phones, computers, network infrastructure, the internet, telecommunications systems, and related services and applications. In recent times, there has been a growing fascination with the potential of computers and the internet to enhance social communication across various levels. This curiosity has driven research and advancements in the realm of ICT and social digital media platform, leading to the emergence of interactive technologies like Web 2.0, often referblack to as participatory technologies as depicted in Fig. 1 (Hu and Yu 2021).

Throughout this analysis, authors took a close look at the research that has been done on how people in different social circles interact with various categories of consumer gadgets. This paper evaluates the conceptual paradigms used in the existing investigation, as well as the dimensions used to assess online platforms, study circumstances, and survey results, in order to determine shortcomings in the literature and potential for future research. The CICR framework was used in the analysis, which is commonly used in clinical studies with strong scientific foundations. Lockwood et al. (2015) say that a modified version of the CICR paradigm, which is based on the traditional CICR, can help researchers come up with clear study hypotheses. The survey questionnaire guided the study's investigation, which was developed using CICR:

1. Which conceptual models are most frequently invoked in attempts to clarify the pblackictive power of cultural connections with regard to consumer technological adoption?
2. Can there be coherence between the conceptualization, design, and implementation of important social connectivity structures?
3. What types of investigation settings were used in the publications examined?
4. Exist commonly studied areas and new tendencies for future research?

A time frame of 10 years between 2010 and 2022 was chosen because the main goal was to focus on the most recent and most important papers. These papers consider changing consumer behavior and social connectivity experiences influenced by recent factors, such as technological advances.

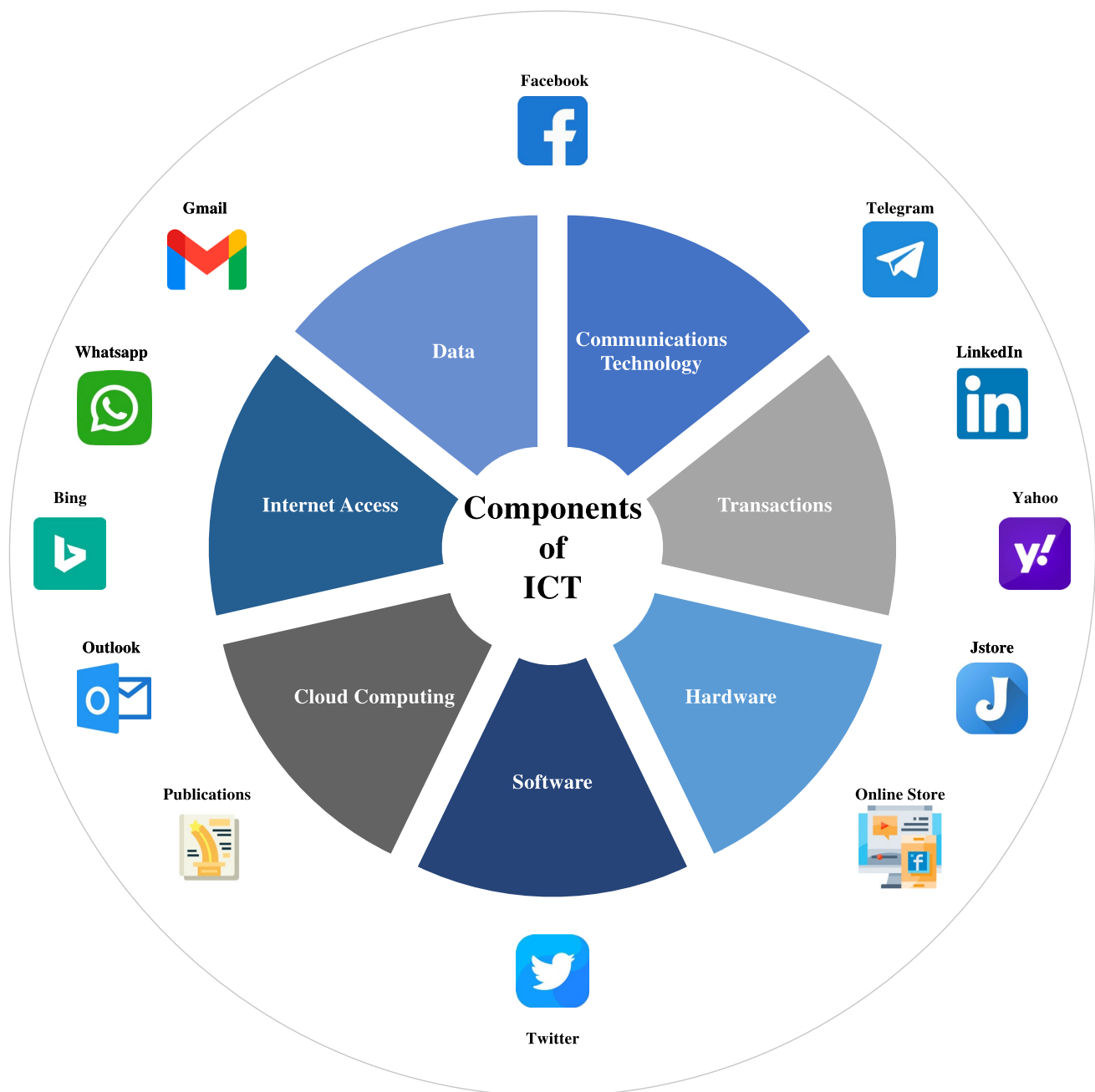


Fig. 1 Social media platform in terms of ICT

Another reason for making this choice was to focus on the many current papers.

The structure of the article is as follows: In Chapter 2, the authors discussed the expanding area of study on social media platforms as a factor influencing technology adoption. The dispersed strategy in the theoretical underpinnings that have been applied is also emphasized. The third chapter offers an overview of the investigation's procedure that was employed to classify, pick, evaluate, and summarize research on social media platforms and consumer information

technology utilization that was accepted for publication within the stated period of the survey. These steps were performed in order to answer the questions posed in the second part. The findings of the 69 publications that were chosen and analyzed are presented in Chapter 4, along with the conclusions that were drawn from the research inquiries that served as a guide. A review of the results makes up the fifth portion of this paper. This article comes to a close with a conclusion and some recommendations for further study in the last part.

2 The influence of social media platforms on consumer technological adoption

Academic pursuit of community connections as a factor in consumer technological adoption has increased. According to Pan (2010), a person's social media platforms affect their confidence, attitudes, and actions. Research has demonstrated that communication of thoughts and ideas during social relationships influences usage patterns (Sheldon 2002; Bilgihan et al. 2016; Siciliano et al. 2017). Nevertheless, the available research demonstrates that the ideological bases, classifications, related concepts, and indicators of social networking were inconsistent, leading to inconsistent and contradictory outcomes that restrict the generalizability of such observations (Meng et al. 2017).

The idea of a social media platform, which comes up a lot in research on networking sites, has been described in different ways in different studies, showing a lack of coherence. In Zhou et al. (2015); Khan et al. (2021) authors figure out how likely it was that a cultural object on a social media platform would support a product, activity, or point of view. The hypothesis of socio-cultural networks and the Social Data Processing Design (SDPD) served as the foundation for this investigation. The research evaluated societal impact, architectural equivalency, object homology, and predisposing variables in order to anticipate acceptance decisions between agents on a social media platform. The investigation's results demonstrated that societal impact had limited prognostic validity and that predisposing variables (undiscovered factors) are crucial in predicting acceptance likelihood.

Authors in Filieri et al. (2014), investigated the link between connection design as well as acceptance assessments, and quantified media platforms using emulation, management, lock-in, likeness, relative frequency, and group impacts. To do so, they made use of the notion of social media networks. Using data from open applications and action linkage panel statistics, the research found that an individual user's willingness to adopt a technique was significantly influenced by either their own activities or those of others in the community with whom they are associated. This was found to be the case regardless of whether the user was involved in the development of the technique or not.

In Okello Candiya Bongomin et al. (2018) authors have examined the mitigating impact of social connections in the association among digital payment usage and access to monetary services in sub-Saharan Africa in research that also utilized social media platform theory. The study examined online communities based on the operational quality of connection linkages as defined by node centralization, network size, resilience, and intertextuality to

describe their impact on the amount of information shared and transmitted within social media platforms. The findings from the research were that the presence of both moderate and strong social media site linkages among electronic payment consumers increased access to monetary services. The methodologies and conclusions of the preceding investigations highlight the lack of key indicators for community networking theory, leading to various indicators and results that make it challenging to apply empirical data.

In the study of how people connect with each other, the idea of "social capital" has also become more popular. Theory of social capital says that faith, traditions, and connections between people are assets Mishra (2020). Authors in Sabatini (2009) used the terms joining capital, spanning capital, and connecting capital to describe how social ties affect how people use new technologies. In Lee and Katz (2015) authors investigate into how Korean immigrants' use of mobile connectivity changed the online communities they were a part of. The research implemented the concepts of constrained cooperation, social networks, and relationship commitment. Utilizing social value, node density, heterogeneity, and centralization, the social media platform concept was operationally defined. The use of linking capital to measure social wealth demonstrated the ability of social data centers (SDCs) to increase connecting social value. Linking capital and bridging capital, however, were not considered in this research. In addition, social capital was employed as a term associated with the social network concept instead of a theoretical model, thus illustrating the current ambiguity in the research on social connections as a significant determinant of consumer tech consumption.

In a study that examined the adoption of innovation by agro-food developers through social media platforms authors used the social value concept to clarify the impact of companionship, connecting, and bringing capital on the dispersion of modern farming among many farming communities (Ramirez et al. 2018). The findings suggest that local social connections exerted a greater effect on the judgment calls, attitudes, and behaviors of their participants. Similarly, in research that utilized the community capital concept to clarify the impact of community capital on destination technology acceptance for touristic promotion researchers quantified the concept utilizing bonding capital as well as connecting capital and stretched it with bandwidth utilization, relationship quality, belief, and social rules (Sung et al. 2015). Several dimensions employed to implement social value in the research imposed a major impact on the degree of technology acceptance within the destination promotion company, the analysis suggested. Authors in Micheels and Nolan (2016) employed social value as a category to evaluate its implications in connection to permeability and the acceptance of farming technologies in Canadian

provinces. The study measures social value by promoting social engagement, trustworthiness, common identity, and participation in neighborhood organizations. This highlights the heterogeneity that occurs in the use of social capital as a concept and occasionally as a component of social networking analysis.

Similarly, the concept of social learning was applied in the investigation of community connections' impact on user technological adoption. The idea asserts that societal connections among individuals on a social media platform promote transformation in cognition outside the scope of the individuals or within broader societal entities (Micheels and Nolan 2016). Individuals' activities in a community are influenced by their network encounters, according to Garip et al. (2020). In their explanation of the use of digital payments on social networking sites, authors contrasted rural and urban Ugandans (Kiconco et al. 2020). The investigation was based on the theory of social learning and operationally defined through monitoring, imitating, hands-on practice, and explicit technical support services provided by social media network players. The investigation found that social networking features helped learning more effectively than personal qualities.

In a different study that used the idea of social learning, authors in Vishnu et al. (2019), investigated how landowners' social media platforms affect how they learn about important cattle equipment. The presence of a competent personality in the group and homogeneous social communities that encouraged the transfer of innovation and creation were both identified as having an impact on social learning. Using the idea of social learning, authors in Murendo et al. (2018) looked at how social media platforms affected how digital payments were used in Uganda. The theory is operationally defined by combining the strong points of the feeble connections concept and the societal assets theory (Lin 2017; Ahmed et al. 2023) with the close link and weaker tie constructions of the theory. According to the data, social group knowledge aided in the spread of information about digital payments and increased their use. Scholars have a hard time using the early theoretical and conceptual structures to study online platforms, as shown by the fact that they use very different methods to implement concepts and metrics for the theory of social learning and try to expand the idea of social learning with structures from other hypotheses.

Previous research is flawed and results to misunderstanding because of the absence of uniformity in the conceptualization as well as operationalization of components and the assessment elements and levels, as evidenced by the preceding debate. Consequently, it is necessary to conduct a comprehensive scholarly literature survey of investigations on social connectivity and consumer innovation utilization in order to determine known vulnerabilities and prevalent

themes that might set the stage for prospective research on simplifying the emphasis on social connectivity as a crucial component in the research on consumer innovation utilization platforms.

3 Approach

To better understand the connection among social media platforms and consumer technological adoption, an in-depth research analysis is carried out in this research study. A systematic analysis of the articles is a well-thought-out way to look at the body of empirical studies that have already been done. This is achieved through standardized, repeatable procedures to identify relevant papers, select the most pertinent studies, critically evaluate them, and report the findings (Lin 2017). Meta-analyses are thorough and comprehensive, and they provide information regarding the procedures and strategies employed in a particular study (Fisch and Block 2018). Also, because they answer certain research questions, they make it possible to draw valid conclusions.

3.1 Planning

The keywords "social media platforms", "technological consumption," and "private individuals" were used during the investigation. Because it has been shown that the terms "technology consumption" and "technology acquisition" are often switched, the phrase "technology integration" was added. Throughout the assessment of authentic identified articles, the study verified that only investigations that examined actual consumer technological consumption were preserved, irrespective of whether the study collection employed technology consumption or technology acquisition.

3.2 Investigation in previous publications

The investigation took place in two distinct phases. The first investigation was executed on Google Scholar by utilizing keywords "social media platforms and innovations", "social media platforms and technology usage between specific consumers", "social media platforms and innovation strategies", "online communities and digital adoption between participant clients", "networking sites and technological advancements utilization", "web 2.0 and innovative technologies utilization between individual end facebook, twitter and new tech acquisition", and, finally, "networking sites and engineering utilization between individual clients." The goal of altering the keywords was to verify that the plan had reached a point of fullness when no more articles could be discovered. It was found that Google Scholar is a good place to start because it gathers academic papers from

different sources. Having Google Scholar, we were able to identify the database systems with the maximum throughput of relevant academic journals by name, which assisted us in identifying the most highly significant and positive sources for an in-depth investigation. The analysis yielded several outcomes with numerous overlapping components. The findings of the analysis are displayed in Table 1.

3.3 Investigate related publications highlighted by Google scholar

The results of the preliminary investigation made it possible to move on to the next stage of the search. It was determined which sources had the greatest amount of academic articles included. The options that were mentioned were JSTOR, Taylor & Francis, Sage, Springer, and Science Direct. Wiley Online was also mentioned. Following the aforementioned keywords, a check was performed across all of the indicated repositories. The below sequence was used to analyze and

classify the findings: Using the survey's title to determine the significance of the survey's findings.

Descriptions were utilized to even further select the published articles that dealt with the connection among social media platforms and technological adoption from those that completed the phase. The analysis rejected any articles that does not use the individual customer as the basis for evaluation. These comprised research with enterprises as the primary unit of investigation. During the screening procedure, only investigations that examined real technology adoption were preserved.

The blackuction of the group of articles to 69 journal articles that satisfied the research goals is depicted in Table 2. Likewise, Fig. 2 depicts the stream of the comprehensive survey's choosing procedure. The PRISMA flowchart illustrating the selection process of manuscripts included in a comprehensive literature survey. There are four key phases in this systematic approach. In the Identification phase, an integrated search was conducted using Google Scholar, which resulted in an initial pool of 364,016 articles. After eliminating duplicate and incorrectly tagged articles, 362,659 were removed, leaving 1357 articles for further consideration based on relevant keywords. During the Screening phase, the abstracts or summaries of the identified articles were examined. This step led to the retention of 295 manuscripts, with 1062 articles excluded for not meeting eligibility requirements. In the Eligibility phase, the full texts of the screened manuscripts were thoroughly reviewed, ultimately shortlisting 88 manuscripts, while 16 were excluded for failing to meet the full-text review criteria. Finally, in the Inclusion phase, 72 manuscripts were selected and included in the literature survey for comprehensive analysis.

The next step was to get information from the publications that had already been looked at and passed the description assessment phase. The authors created an outline that included the following sections: (I) the title of the published article, the researcher(s), the date it was published, and the publishing company; (II) the purpose

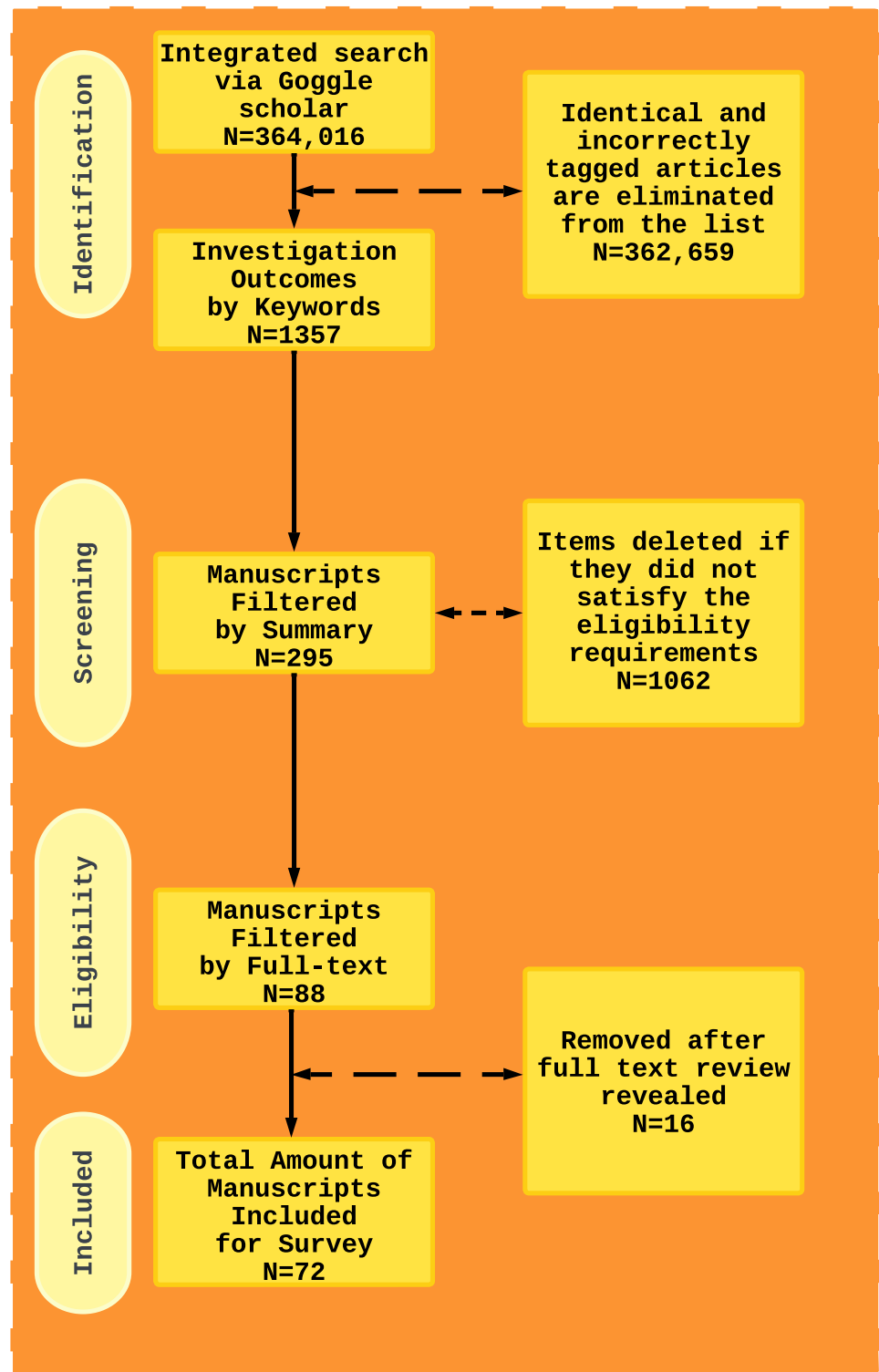
Table 1 Google scholar results for specific keywords

Keywords	Outcome	Time (Seconds)
Social media platforms and innovations	930,000	0.24
Social media platforms and technology usage between specific consumers	381,000	0.35
Social media platforms and innovation strategies	1,370,000	0.25
Online communities and digital adoption between participant clients	80,600	0.22
Networking sites and technological advancements utilization	40,700	0.24
Web 2.0 and innovative technologies utilization between individual	114,000	0.44
Facebook+ and Twitter and New tech acquisition	550,000	0.19
Networking sites and engineering utilization between individual clients	72,100	0.39

Table 2 Summary of keywords and outcomes

Investigated databases	Investigation outcomes by keywords	Manuscripts filtered using the titles of the studies	Manuscripts filtered by summary	Manuscripts filtered by full-text	Total amount of manuscripts included for survey
Springer nature Group	1601	21	11	9	8
Elsevier	69,304	251	51	17	16
Sage publication	91,233	211	55	11	9
Jstore	31,322	270	54	18	9
Wiley library Online	87,254	93	33	13	11
Taylor and Francis	83,254	511	91	20	19
Total	364,016	1,357	295	88	72

Fig. 2 PRISMA flowchart for the comprehensive literature survey's choice procedure



of the study; (III) the study background; (IV) a description of social media networks and research structures; (V) a description of technological utilization and investigation

frameworks; (VI) theoretical underpinnings; (VII) the conceptual model used; (VIII) observations; and (IX) research

problems identified. The authors then extracted and compiled the information using this framework.

4 Survey outcomes

Survey question 1: What popular conceptual paradigms are applied to clarify social media platforms as a determinant of consumer tech adoption?

4.1 Disintegration of conceptual paradigms

This was evident from publications that identified social media platforms as a concept underpinned by diverse

theories, each with distinct axioms, definitions, and measurable variables. Splitting of theoretical paradigms led to contradictory results and a difficulty extrapolating the results. In literature studies to understand consumer technological adoption, behavioral research and data system concepts, such as data dissemination and the integrated concept of adoption, were used (Ramdhani et al. 2014; Dwivedi et al. 2019; Lai 2017; Venkatesh et al. 2012; Niehaves and Plattfaut 2014; Pai and Huang 2011; Wen et al. 2011; Turel et al. 2011). The various hypotheses are summarized in Table 3 to illustrate the fragmented technique. Survey illustrates the variety of ideas employed to describe social media platforms, as well as the range of theoretical hypotheses established in the research listed in Table 4.

Table 3 Frequency of investigations across various literature settings in social media platform studies

Literature setting	Frequency of investigations	Literature setting	Frequency of investigations
Utilization social media platforms	30	Work efficiency	1
Pedagogy	5	Catastrophe recuperation	1
Entrepreneurship	4	Hospitality	1
Marketing	1	Consumer contentment	1
Migrants	3	Societal effort	1
Consumer conduct	2	Security	1
Cell phone consumption	5	Farming	1
Medicare	1	Ecological development	6
Community interaction	2	Design and development	3
Civic engagement	1	Networking	2
Total		72	

Table 4 Concepts and theories in social media platforms: A comparative overview of publication frequencies

Concept	Quantity of publications employing the concept	Concept	Quantity of publications employing the concept
Social media platform theories			
Social media concept	14	Theory of organizational polycentricism	1
Social value concept	10	Integrated theory of adoption	3
Data dispersion concept	8	Directly effect hypothesis	1
Theories of socialization	6	Resource-based perspective	3
Online communities research	5	Social connectivity concept	4
Framework of technological acceptance	5	Theory of bounded rationality	1
Theory of societal impact	5	Gratification theory	1
Pattern of social cues acquisition	4	Grounded theory	1
Pattern of elaboration probability	4	Service quality model	1
Theory of relationship commitment	2	Affect-as information theory	1
Public role theory	2	Resource based view	1
Theory of pblacketermined conduct	2	Social role theory	1

5 Survey question 2: Is the operational definition of social media platform components consistent?

The analysis of the academic studies suggests that several theoretical models and categories were employed by previous researchers to substantiate the links that were examined. Sun et al. (2014) utilized social media platform concept and examined the importance of lock-in impact, copying impact, resemblance and authority impacts in a study that investigated the connection among social media sites on the web and technological acceptance. In a research of the interplay among social platforms and protest movements, Jost, Barberá, Bonneau, Langer, Metzger, Nagler, Sterling, Jost et al. (2018), quantified online platforms utilizing sharing of information, inspirational material, and synchronization of activism operations as the research builds, whereas, in an investigation into the connection among networking site connections and talent acquisition, Nayak, Bhatnagar, Bhatnagar and Budhwar (2018), established the Jia (2018), utilized the Bass modelling approach and emphasized on the intensity of connection linkages to implement social media networks employing range value, intimacy, and centralization as the study concepts.

Previous study's tendency to examine various dimensions and their interrelationships resulted in an upsurge in the segmentation of results, which made it hard to apply them. Intriguingly, there have been limited academic initiatives to incorporate these past research efforts in order to provide an implementation strategy that'd facilitate the generalization of results about the linkages among social media platform and consumer technological consumption variables.

6 Survey question 3: What investigation settings did the examined research employ?

According to the examined publications, there is a massive rise in the role played by social connections in affecting customer technology adoption. Frequent studies settings included schooling, entrepreneur, security, employability, farming, and medical. The study of how to utilize technology to enhance social connections through social media platforms has also arisen as a topic of serious importance.

Survey question 4: Are there recurring themes or directions in the studied literature that might inform prospective research?

According to the findings of the survey, three distinct research directions have emerged.

6.1 First research direction: the impact of social communication connections on consumer technological consumption

The first direction addressed social connections from a functional standpoint, focusing on how the architecture of societal connections affected consumer technological consumption and mostly relying on social networking modeling theories. Social networking analytics (SNA) is a conceptual paradigm that conceptualizes the architecture of social systems by concentrating on the features of the links linking individuals as opposed to the personal qualities of the members. SNA employs numerical frameworks to determine the functional relationships between endpoints in a social media platform, as well as how their interconnection affects behavior, by utilizing concepts such as extent of primacy, adjacency, connectivity primacy, eigenvalues indispensability, systemic congruence, and connectivity primacy (Ahmed et al. 2022, 2023).

Ludlow and Heydari (2014) used quantitative computational approaches to evaluate the effect of amount of centralization and adjacency within social media platforms in a study on the method for selecting prominent people for promoting a novel item using social media platforms. The research revealed that social connectivity links with the greatest level of centralization and exerts the most influence over item acceptance. Ludlow and Heydari (2014), investigated how social media platforms impacted users' use of mobile technologies for cooperation. In a regulated field, research involving 327 individuals, networking site analysis to quantify eigenvalues of centralization and connectivity friendliness between all of a social network's components. Eigenvector centralization evaluates the effectiveness of an unit while taking into account the significance of its neighbors. The results suggest that adding social media platforms to a mobile computing system can make it easier for nodes to share information and learn more.

Hinz et al. (2014) examined the impact of social media platforms on buyer choices in a separate study including 300 respondents. Utilizing amount of centralization (the amount of social relationships a client has inside a connection) between all hubs in the research, analytical graph theory was employed, and it was determined that architectural comparability influenced acceptance of technology.

6.2 Second research direction: The impact of social networking connections on consumer technological consumption

The second line of investigation focused on examining the influence of network linkages and actor connections on the utilisation of consumer technology. The concepts of social networks, societal learning, societal capital, and

link strength are widely recognised in academic discourse. Partnerships (homophily) were commonly characterised based on several criteria, including the degree of closeness, amount of contact, trustworthiness, and level of likeness. Based on the results of inquiries examining the composition of connections within online social networking platforms, it was observed that the dissemination of perspectives and viewpoints during interpersonal interactions had an impact on patterns of usage Hossain and de Silva (2009).

6.3 Third research direction: Social networking platforms and its impact on consumer technological consumption

The third category indicated a growing and dominant investigation into social connections made possible by the proliferation of social media web pages (SMW). The stream also addressed the third survey question regarding developing patterns. Investigating how individuals utilise social media platforms to connect, share information, and affect one another's decision-making is a phenomenon that is gaining traction in academic studies. The evaluated research revealed an increasing focus in research on the influence technology has had on social media platforms as a result of the rise of technology-driven SMW. The investigations highlighted the degree to which digital has been ingrained in customers' daily lives and the impact of social media platforms on consumer technological consumption.

This line of research revealed the same heterogeneity of conceptual approaches, categorization of notions, and results as prior studies assessed for this research. Katona et al. (2011); Alvi et al. (2022) utilised the diffusion concept and evaluated extent impact, density of links, and adjacency; whereas they utilised the socialisation concept and evaluated system usage, social involvement, perceptions of the quality, and notoriety; and Peng and Mu (2011) utilised the social media network concept and evaluated lock-in effect, mimicking impact, resemblance impact, leadership impact, and recency effect. The outcomes were as diverse as the theoretical perspectives and notions employed in the several investigations.

Zubcsek et al. (2017) observed that as influencing authority grows, more and more individuals possess a "brokering" position between their connections, and that the level and grouping impacts increased customers' acceptance of technology. The Mu et al. (2017b), discovered that an user's choice to employ a particular technique was greatly impacted by the responses that others had in their online communication collective and deduced that the sharing of information inside a social media platform was effective.

7 Discussion

The goal of this study was to do a thorough analysis of the current research on the relationship between social media platforms and how people use new consumer innovations. For this analysis, a maximum of 72 empirical investigations were chosen. Using the investigation topics as a foundation, a systematic literature survey was carried out. The survey demonstrated an increasing demand for empirical research on the connection between online communities and consumer technological consumption. Many theoretical paradigms are employed to describe the connection between social networking and consumer technology adoption, which has led to dispersion in the increasing number of studies. The factors employed, operationalization, and assessment of structures, as well as the conclusions, indicated heterogeneity, making it impossible to generalise the conclusions of the many studies evaluated. In addition, this constraint impedes the implementation of the data by professionals who wish to leverage the insights in product design and marketing activities utilising social media platforms as a significant factor in customer technology adoption.

8 Description, theoretisation, and utilization of social media platforms

The following sections describes the social media platforms and their relationship with consumer technology adoption. This section include a description of social media platforms, the theoretical frameworks employed to understand them, and the connection between platform utilization and consumer technology usage.

8.1 Description of social media platforms

There were indications of agreement in the studied material about the description of social media platforms. Social media platforms can be considered to be composed of individuals or groups. Players on social media platforms are associated via connections and linkages that promote interaction and trade, through whom they seek out knowledge that impacts their conduct (Putranto et al. 2023; Abbas et al. 2022; Ramirez et al. 2018).

8.2 Theoretisation of social media platforms

The lack of agreement on theories that clarify the social media network concept was largely responsible for the variation in definitions of social connections. As indicated, several studies have incorporated diverse ideologies pertinent

to their fields of research. This approach's inconsistency exposed confusion, which resulted in diverse results and disintegration. Nonetheless, the increased fascination with this area of research and the discoverable segmentation imply that social networks are increasingly recognised as a key determinant of consumer technology adoption. Therefore, it is an essential necessity to consolidate this subject matter by constructing logical and analytical paradigms that will direct the development of prospective studies.

8.3 Connection among social media platform utilization and consumer technologies

A study of the existing research literature found that many researchers' perspectives on the importance of social connections in describing the spectrum of consumer tech utilization converged.

8.4 Definitions of utilization of consumer technologies

Researchers haven't yet agreed on a single definition of "consumer technology usage," according to the reviewed studies. The researchers in Oyman et al. (2022) stated that consumer technology usage was indeed the end of the procedure in which consumers made choices about a device, and that usage occurs whenever a workgroup demonstrated readiness to utilise the technology for the reason that it was intended for. In contrast, Alshurideh et al. (2023) argued that consumer technology adoption rates are a vital measurement of a data system's success. The use of consumer technologies has been called "technology adoption" by some academics. For instance, Putranto et al. (2023) described it as the time it took for a consumer to go from learning about an invention to developing an opinion towards it to deciding whether or not to adopt the technology.

According to existing knowledge, the employment of consumer technology by individuals, teams, and enterprises is a central concept in the study of data networks Hoffmann and Akbar (2023). As per Usman et al. (2022), nevertheless, there is a large variety of system consumption measurements, which makes it hard for researchers to evaluate distinct research outcomes. Two categories of system utilization measurements exist: objective computer-recorded metrics and subjectively self-reported metrics (Pan 2010). The literature review revealed fragmentation caused by the multiple theoretical underpinnings employed and the situational perspective on the definition, categorization, and assessment of consumer tech consumption. Yet, the interpretations shackle the recognition of the effect of social connections and personal and collective views on consumer conceptions about technology adoption.

Research shows that the way users, teams, and businesses use consumer technologies is a key part of the study of data platforms. But it has a lot of different system usage measurements, which makes it hard for researchers to compare different study results. Two categories of system utilization measurements exist: objectively computer-recorded metrics and subjectively self-reported metrics. The analysis of the literature revealed heterogeneity brought about by the use of various conceptual frameworks and the situational method of identifying, interpreting, and evaluating consumer technology consumption. However, there was a common theme among the classifications: the recognition of the influence that social connections and individual and group viewpoints have on consumers' perceptions of technology adoption.

9 Positive impact of social community forums on consumers adoption of modern digital technologies

The advent of the technological age has brought about significant transformations in various aspects of human existence, including lifestyle, employment, and social interactions. The incorporation of mobile devices and intelligent technology into our everyday routines has become indispensable. However, the process by which customers accept and embrace these technological advances remains unclear. Social community forums have come to serve as a noteworthy mediator for the adoption of technological advances. These digital platforms afford people of all ages the chance to establish connections, exchange information, and engage in cooperative endeavours with people who possess similar interests and goals. This section examines the favourable effects of social community forums on the implementation of contemporary technological advances by consumers. It emphasises the manner in which these platforms enable individuals and cultivate breakthroughs (Gautier and Fikru 2022).

9.1 Improved information sharing

Social community forums function as intellectual epicentres where people can acquire erudition and engage in the sharing of erudite discourse pertaining to contemporary technological advances. These platforms serve as a nexus for individuals who possess fervent passion, profound expertise, and a proclivity for embracing nascent technologies. These individuals voluntarily engage in the dissemination of their personal encounters, astute recommendations, and optimal methodologies. By actively engaging in debates and communications, users acquire essential knowledge, direction, and suggestions. Through the acquisition and assimilation of said knowledge, individuals are endowed with a heightened level of awareness and understanding pertaining to the

advantageous attributes, distinctive characteristics, and prospective utilities of diverse technological innovations Abbas et al. (2022).

9.2 Developing trust and confronting challenges

The process of integrating new innovations frequently entails encountering various obstacles and concerns. Social community forums serve a crucial role in mitigating these impediments by cultivating trust between consumers. Those who exhibit hesitancy or uncertainty towards the adoption of an emerging technology may take part in seeking guidance, posing inquiries, and obtaining assurance from fellow participants within online forums who have gone through the effective implementation of said technologies. This communication fosters a sense of assurance and aids users in overcoming any uncertainties or apprehensions they may have. In addition, forums offer a medium for facilitating open dialogues regarding the possible obstacles and constraints linked to specific innovations. Individuals have the ability to freely express their own life experiences, encompassing both favourable and unfavourable aspects, thereby empowering other individuals to make intelligent decisions. The establishment of transparency fosters confidence as well as accountability throughout the community, thereby promoting the acceptance and utilisation of technology by consumers Haddad et al. (2022).

9.3 Assistance and problem solving

Contemporary technological innovations occasionally exhibit intricacy, thereby presenting consumers with potential challenges pertaining to technical malfunctions or complications during the process of installation and utilisation. Social community forums are highly effective in offering an appropriate setting for people to seek support and resolve issues. The combined skills and expertise of participants in online forums enable users to promptly find remedies to their inquiries and problems (Bayer et al. 2022). Social community forums have emerged as influential drivers in the process of consumers embracing contemporary technological advances. These networks offer avenues for the exchange of knowledge, the establishment of trust, surmounting obstacles, acquiring assistance, and promoting creativity. By harnessing the shablack expertise of people who share similar perspectives, consumers are able to make well-informed choices, effectively navigate obstacles, and fully embrace the possibilities offered by these advances in technology. The ongoing progression of digital advancements has led to the significant impact of social community forums in influencing the trajectory of digital implementation, facilitating consumer empowerment, and nurturing a digitally interconnected society.

10 Exploring challenges and foundations in social media platform research

This sections address themes related to social media platforms and their impact on consumer technology adoption. The negative impact of social community forums examines the drawbacks, including misinformation, unhealthy discourse, and overreliance on subjective opinions, which hinder effective adoption of modern digital technologies. The pillars of theory for social media platforms highlight the lack of unified theoretical frameworks, advocating for a cognitive paradigm to empirically explore the relationship between social connections and technology use. Finally, the implementation of social media platforms discusses the varied frameworks used to operationalize these platforms, emphasizing the inconsistency in findings and the need for a standardized assessment approach. These sections collectively aim to provide a comprehensive understanding of the challenges and opportunities in this domain.

10.1 Negative impact of social community forums on consumers adoption of modern digital technologies

The adoption of internet-based online communities has unquestionably exerted a substantial influence on the proliferation of contemporary technological advances. However, it is imperative to acknowledge that these networks can also have adverse consequences for the public. One notable drawback is the possibility of disseminating inaccurate information and subjective viewpoints Al-Emran and Griffy-Brown (2023). Given the diverse range of participants within online forums, users often encounter difficulties in discerning reliable information amidst the presence of subjective viewpoints, individual prejudices, and the potentially intentional dissemination of false or misleading content. The potential outcome of disarray and uninformed decision-making can impede the effective implementation of technological advances. In addition, online forums have the potential to foster an environment conducive to negativity, as users may partake in contentious discussions or disseminate unjustified critique. The presence of an unhealthy atmosphere may have a detrimental effect on people's readiness to engage actively or seek support, thereby hindering the broad acceptance of technological advances Adke et al. (2022); Ahmed et al. (2022). Furthermore, it is possible that social networking forums unintentionally cultivate a tendency to excessively depend on user views instead of engaging in thorough studies or seeking advice from professionals. This phenomenon can lead to users making uninformed decisions or disregarding

other innovations that might be more suitable for their requirements. Hence, it is imperative for individuals to approach public forums with prudence, employing analytical skills, and striving for a comprehensive viewpoint in order to make informed choices regarding the integration of contemporary technological advances Dhiman (2023).

10.2 Pillars of theory for social media platforms

The comprehensive assessment of the research observed the absence of a major idea or theories with reliable as well as empirically validated components and assessment elements that might be utilized to describe the impact of social media platforms on consumer technological consumption. In addition, an overview of the underlying assumptions shablack by multiple theories is presented. Evidently, future research must concentrate on building a cognitive paradigm to underpin empirical investigations on social media platforms and technological utilization. Such a structure would facilitate the development of theoretical underpinnings in the investigation of social connections. It would additionally open up opportunities for subsequent research that develops new concepts that promote the increased curiosity about the investigation of the link between social connections and consumer technological use.

10.3 Implementation of social media platforms

To operationalize social media platforms, various frameworks were utilized. In the absence of a benchmark assessment tool, the results of the investigations were inconsistent and contradictory Rezaei et al. (2019). For example, both Alshurideh et al. (2023) and Putranto et al. (2023) utilized the social value concept but with different definitions. Abbas et al. (2022) discovered, in contradiction to Hoffmann and Akbar (2023), that social media platform links strongly impacted consumer technological consumption. They contended that close connections had a greater impact on consumer technological advances consumption than feeble linkages, whereas Baierl et al. (2022) discovered that weak ties had a greater impact on consumer innovation consumption than deep links, as well as Ninaus et al. (2021) contended that bridging ties/brokering.

11 Tech-driven evolution of social media platforms

The analysis showed that people are getting more and more interested in how technology is making social networking grow. The world has become more interconnected due to the near-ubiquity of Internet access. With the development of Web 2.0, imaginary societies have formed where

individuals may communicate, network, voice their opinions, and exert power over everyone else. With online word-of-mouth (WOM), users of an internet group (such as social media platforms) can express their ideas and evaluations about a particular product, issue, or group for free, which has a substantial impact on consumers' buying inclinations. Thus, social relationships are currently a significant element of consumers' online activity, with over 1.5 billion people sharing thoughts, experiences, evaluations, and images with their internet platforms.

The analysis showed that people By the year 2022, sixty percent of the total residents of the globe has access to the internet. Because of the widespread adoption of the internet, social networking sites (SNS) have developed into a massive force, especially in influencing the communications and choice behaviors of customers. Users have gone past utilizing SNA for social and recreational life to making major use of it in medical services, administration, and industry, which has resulted in a transformation in the nature of relationships among the worldwide population. According to the findings of one survey, over ninety of medical professionals use digital platforms for their personal lives, whereas only sixty-five percent use it for their official lives are getting more and more interested in how technology is making social networking grow. The world has become more interconnected due to the near-ubiquity of Internet access. With the development of Web 2.0, imaginary societies have formed where individuals may communicate, network, voice their opinions, and exert power over everyone else. With online word-of-mouth (WOM), users of an internet group (such as social media platforms) can express their ideas and evaluations about a particular product, issue, or group for free, which has a substantial impact on consumers' buying inclinations. Thus, social relationships are currently a significant element of consumers' online activity, with over 1.5 billion people sharing thoughts, experiences, evaluations, and images with their internet platforms.

According to the results of the analysis, the development of technology has changed a lot about how people interact with each other, talk to each other, look for information, get help, and work together to achieve goals. Network effects are a tremendous advantage that the internet has given to the social relationships that people have. The term "network externality" refers to a rise in the value of a good or service that is caused by a rise in the total number of individuals who use a product that is functionally equivalent to the one in question. The network externalities influence the technologies that are adopted and deployed. Members of an internet-based social network are brought together by shablack interests in issues such as medicine, unemployment, housing, faith, school, and the environment, as well as political activity. Teaching and learning are both supported by social media platforms as well. According to published research,

learning networks and the distribution and exchange of information resources by collaborative project parties have recently emerged.

11.1 Social media platform's as developer of social capital

Social capital has been characterized by academics as the assets that are accessible to people and organizations that are components of a social network Boruzie et al. (2022); Warschauer (2003). According to Katebi et al. (2022), the participants in such relationships are bound together by a shablack trustworthiness and values. Bonding wealth, linking capital, and linking investment are the three distinct kinds of social capital that were defined by Usman et al. (2022). At the exact same time, Santos et al. (2022) characterized social value as the benevolence that is created by the web of social ties and that may be deployed to assist action. In other words, social capital is an asset that can be used to assist action. Although the approaches taken by various academics to defining social capital are not identical, they do believe in certain aspects of the concept. For example, they all recognize that social capital is comprised of organized networks among people of all kinds, as well as the assets that accumulate from connections among the stakeholders. The researchers are also in agreement that social capital makes it easier for individuals to take action within social networks.

According to the research that was looked at, the idea of social capital is gaining more and more traction in the social science literature (Tharkude 2023), and it has been regarded as an advantage on social media platforms. The research was informed by social capital on topics including households, youngsters, actions, training, and education, population health, civic life, the democratic republic, leadership, financial growth, and broader problems of mass bargaining, professional success, getting employment, new product development, and supplier involvement, as well as high turnover.

11.2 Findings, research implications, and prospects for further investigation

According to the findings of the 72 lature articles that were examined, it was clear that during the period of time, the investigation into the connection among social media platforms and consumer innovation utilization was becoming prevalent in the fields of entrepreneurialism, schooling, community interactions, advertising, universal healthcare, work performance, and agricultural production. On the other hand, the results showed that the research into the link between social networking sites and mobile communications technology was not enough. This means that more research needs to be done in this area. There was clear evidence of

disjointedness in the way research was conducted, beginning with the selection of theoretical models and continuing through ideation, the operational definition of categories, the measurement of variables, and the presentation of research findings.

11.3 Survey implications

The results have important management and empirical ramifications. Empirically, the study highlights the necessity for further investigation into the establishment of shablack theoretical models that would allow for the extrapolation of study results. Managerial views into the function of societal networking in promoting consumer tech utilization across settings, including cultural characteristics, technological variety, geographic diversity, and others that regulate consumer perception toward innovation and consumer technology utilization, will require additional investigation.

12 Case study

TikTok is a platform that facilitates the development of consumer technological innovations through the utilisation of short-form video information.

12.1 About company

TikTok, a burgeoning social media system under the ownership of ByteDance, has swiftly garnered immense global acclaim and recognition. The app's distinctive value proposition centblack around concise video content has engrossed a vast multitude of consumers, rendering it a compelling subject for scholarly examination regarding the assimilation of modern consumer technology.

12.2 Aims of case study

The primary objective of the present investigation is to delve into the manner in which TikTok has facilitated consumer adaptation to technology by virtue of its inventive strategy pertaining to short-form videos. Furthermore, it seeks to comprehend the ramifications of such a strategy on customer satisfaction and the expansion of the platform. This case study also represents the constructs identified in the systematic review. Which provides a practical demonstration of how community dynamics, relational capital, and perceived value contribute to consumer technology adoption. The platform supports community interactions to build trust and foster reciprocity among users along with aligning with the principles of relational capital from SET. TikTok's personalized algorithms and user-friendly tools enhance

perceived ease of use and perceived usefulness consistent with the constructs from the TAM.

13 Prospective studies directions

13.1 Framework

1. *Campus interviews* A comprehensive questionnaire was undertaken on campus, including students and staff, encompassing a stratified group of 100 individuals who actively engage with the TikTok platform. This group was meticulously selected to ensure a representative cross-section of the population, encompassing a wide spectrum of demographic characteristics. The questionnaire encompassed inquiries pertaining to user conduct, underlying drives, and cognitive interpretations of the platform.
2. *Analysing the data* The information collected from the respondents was subjected to quantitative evaluation in order to discover emerging patterns and trends pertaining to user adaptation to technological innovations, levels of participation, and perceived advantages.

13.2 Outcomes

13.2.1 Technological adaptation

Figure 3 illustrates the levels of familiarity and adaptation among TikTok users. Following are the adaptation ratios;

1. Approximately 85% of TikTok consumers have successfully achieved full adaptation to the platform.
2. The proportion of consumers who have demonstrated a partial adaptation to the app is 10%.

Fig. 3 Ratio of adopting of Toktok platform by users

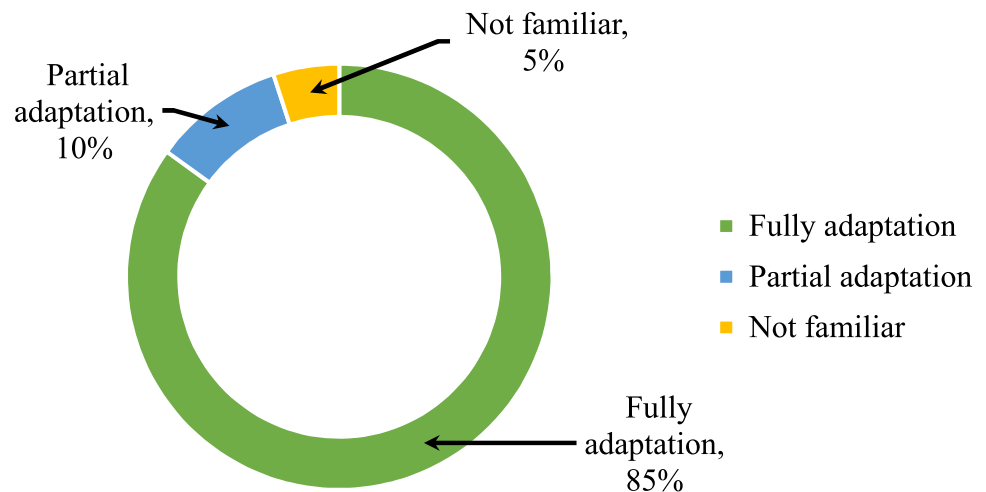


Table 5 TikTok usage statistics

Statistic	Value
Mean daily duration of TikTok usage	Between 1 and 2 h
Proportion of individuals engaging multiple times daily	92%
Mean number of content viewed daily	30

3. The proportion of consumers who are still not familiar to the platform is 5%.

13.2.2 Consumer participation

Table 5 listed the statistics on TikTok usage patterns with average daily duration, user engagement frequency, and content consumption. Stats are as follows,

1. The mean daily duration of TikTok usage ranges from between one and two hours.
2. The proportion of individuals who engage with TikTok on numerous occasions within a single day amounts to 92%.
3. The mean number of content viewed by each consumers on a daily basis is 30.

13.3 Implications for engineering practice

The results derived from this specific case investigation offer valuable insights that can inform engineering practitioners in their efforts to create and build social media systems. Scientists ought to prioritise the development of captivating and habit-forming types of content, utilising personalised algorithms to augment user engagement, and furnishing

user-friendly tools for content generation and communication. Moreover, the cultivation of a dynamic customer base and the establishment of an influencer society can serve as catalysts for enhancing user adoption and facilitating the expansion of the platform. The results highlighted difficulties while also presenting prospects for streamlining research on social connections and consumer tech utilization, two areas that are attracting a growing proportion of scholarly attention as tech keeps governing humanity.

The systematic review emphasized the role of social media platforms in shaping consumer behavior through mechanisms such as trust, reciprocity, and engagement. Here the TikTok platform effectively operationalizes by enabling seamless information exchange and fostering a sense of community among users. The TikTok platform's capacity to encourage user generated content and active participation illustrates the importance of community dynamics as outlined in the CICR framework. TikTok integrates relational capital and perceived value into its strategies through features like short-form videos, interactive content, and influencer networks, driving consumer engagement and technology adoption. Figure 4 demonstrates the levels of TikTok adaptation among participants. Approximately 60% of TikTok users adopt the platform primarily for entertainment and fun; 25% uses it as a creative outlet for producing videos; 10% use it for advertising and brand promotion, and 5% engage for education or professional networking. These diverse use cases demonstrate TikTok's alignment with the proposed unified conceptual framework emphasizing how platforms that address various user needs can effectively foster community, enhance perceived value, and drive technology acceptance.

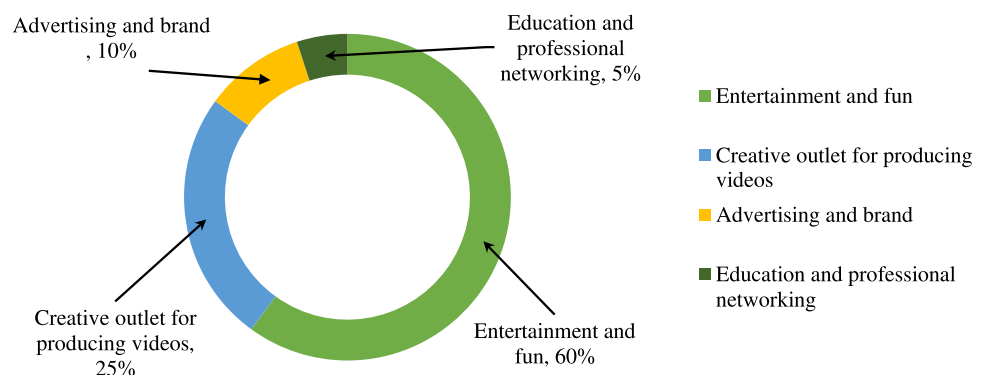
13.4 Design of a theoretical model

Future research could focus on building a shablack theoretical model that includes the consistent assumptions that have been found, with the goal of coming up with a standard theory and way to measure variables for social media platforms. This possible area of future research would

help make standard scale items for social media platforms and lay the groundwork for using empirical data on social media platforms to pblackict how much tech consumers use. Social exchange theory (SET) was not utilized in the various assessed investigations. Behavioral researchers created SET to describe social exchange relationships within social networking sites in light of the reward concept. The empirical analysis of SET is judged by trade regulations and standards, assets transferblack in a social contact, and connections formed as a result of specific interpersonal interaction. SET has an organizing principle applicable to the research in numerous fields. Future studies could evaluate the feasibility of SET as the underlying theoretical paradigm for the investigation of the interaction among social networking and consumer tech use, as well as methods for strengthening its estimates for the future.

In the future, researchers might also look into how ideas about information management could be combined with a behavioral perspective to give a theoretical view of the link between social relationships and consumer tech use. The effect of social media platforms with technology on how people think about technology and how they use it As people use technology more and more in all parts of their lives, normal face-to-face interactions are giving way to connections made possible by technology on the ever-growing number of social media platforms. More research on how people communicate on social networking sites and how that affects how people use technology will help lawmakers and social media networks understand how people really use technology. As technology persists in permeating daily life, engagement, knowledge transfer, and influence between players in social networking sites, particularly social media platforms, will emerge as significant events in consumer technology consumption. Future investigation could concentrate on ways to focus on providing decision makers, institutions, marketing companies, the service industry, and revolutionary activities, among many other priority areas, with data and perspectives on how social media network interplay by consumers affects customer perceptions and consumption patterns in various

Fig. 4 Levels of TikTok adaptation among participants



contexts, such as geographic location, cultural, ethnic, and population ones.

User demographic factors play a dampening impact. According to studies, demographic characteristics comprising age, education, profession, and sexuality influence patterns of behavior, especially adoption decisions. In contrast, she Rola-Rubzen et al. (2020) asserted that women on social media platforms were more inclined to embrace innovation because they benefited from social capital inside these networks. This region offers a chance to do additional empirical research.

13.5 The moderating function of consumer mental traits

The research demonstrates that consumer behavior buffers the connection of social networking linkages as well as consumer tech use. As commonly established by the technology adoption paradigm, consumer technology utilization is significantly pblackicted by consumer behavior, which is evaluated by customer perception regarding technological influences. This field of study is established, but consumer behavior's role as a mediator among social media platforms and consumer tech use has not been extensively investigated. Researchers have shown that buyer decisions regarding technology adoption are frequently made in collaboration with other people or organizations, so this assertion provides a chance to investigate the mediator function of the consumer's behavioral characteristics.

13.6 Limitations and unified conceptual framework

This research study offers important insights into the relationship between social media platforms and consumer adoption of digital technologies. However, authors have acknowledged multiple limitations to provide a balanced perspective and guide future research. The selection of articles included in the review may be subject to publication bias, as studies with significant findings are more likely to be published and accessible, potentially overrepresenting certain perspectives. To address this, meta-analytic techniques to quantitatively evaluate the findings of existing research and mitigate biases should be adopted. The reliance on specific keywords and databases might have excluded relevant studies with alternative terminologies or less accessible indexing. Furthermore, the diverse theoretical frameworks and operational definitions in the included studies posed challenges in synthesizing findings, limiting consistent conclusions. Temporal constraints, such as focusing on studies from 2010 to 2022, may have unnoticed earlier foundational research or recent developments, while regional biases in the analyzed studies could restrict generalizability across cultural or socio-economic contexts. This study's dependence

on secondary data from published literature inherently limits the analysis to existing findings, with variations in quality impacting robustness. Lastly, while the CICR framework structublack the analysis effectively, its novel application to this domain may have missed critical dimensions of the interplay between social media and consumer technology adoption.

Future research can focus on developing a unified conceptual framework that integrates the CICR model with key elements from SET and the TAM to address the theoretical fragmentation in studies on consumer technology adoption. By incorporating SET constructs, such as trust and reciprocity, as moderators, and TAM constructs, such as perceived ease of use and perceived usefulness, as mediators, this framework provides a comprehensive approach to understanding the interplay between social media platforms and technology adoption. This unified model not only offers consistency in operational definitions and measurement but also highlights the importance of community dynamics, relational capital, and perceived value in shaping adoption outcomes. Empirical validation of this framework in diverse contexts, including cultural and demographic variations, can further enhance its applicability and relevance, providing actionable insights for both academia and industry.

In Fig. 5 a unified conceptual framework is proposed for future directives. The framework consists of four core components: Community Dynamics followed by CICR, focusing on the structure, size, and dynamics of online communities that foster interactions, information exchange, and trust-building; Relational capital from SET emphasizing trust and reciprocity among users as moderators that enhance technology adoption by cultivating a sense of belonging and confidence in shablack knowledge; Perceived value from TAM, where perceived usefulness (PU) and perceived ease of use (PEOU) serve as mediators influencing users' decisions to adopt technology by emphasizing its practical utility and integration ease; and Outcome measures capturing dependent variables like increased engagement, sustained use, or innovation diffusion. This figure illustrates the relationships between the CICR framework, SET constructs, TAM constructs, and their influence on community dynamics, relational capital, perceived value, and adoption outcomes.

14 Conclusion

Undoubtedly, the influence of online platforms on the adoption of novel technology by consumers is a significant phenomenon. This particular instrumentality holds significant potential for augmenting cognizance, expediting the exchange of knowledge and ideas, and fostering communal validation. Social media platforms have emerged as a significant catalyst for molding consumer opinions and behaviours

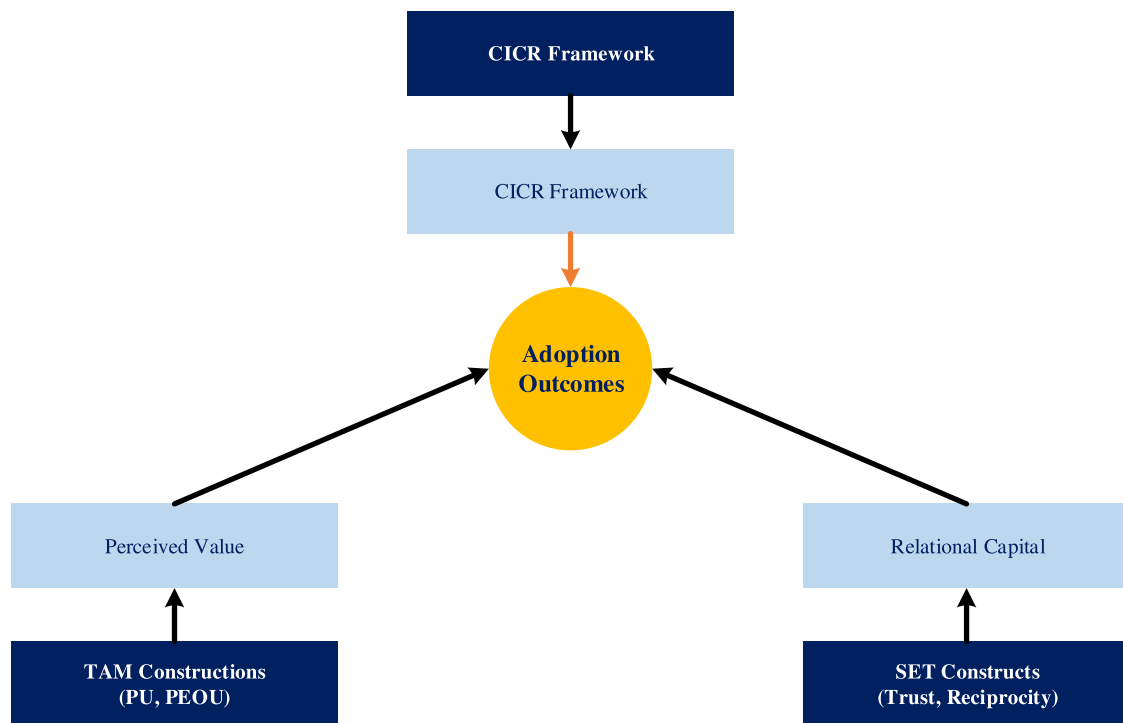


Fig. 5 Unified conceptual framework for future directions

towards novel technologies, owing to their ability to leverage influencer outreach, peer recommendations, and real-time involvement. Given the ever-evolving nature of social platforms, their impact on the adoption of new technologies is poised to expand significantly. As such, it is imperative for enterprises to adeptly harness this platform to facilitate the uptake of their advancements among consumers.

Looking towards the coming years, the potential impact of social media on consumers' adoption of new technology is significant. The proliferation of technology has given rise to numerous intriguing prospects. Tailorblock suggestions for new technologies, which will influence consumer decision-making processes, will be provided by personalised recommendations that are powered by artificial intelligence and data analytics. The integration of augmented reality (AR) and virtual reality (VR) experiences into social media platforms is expected to enhance user engagement by providing immersive opportunities to explore and interact with novel technologies, thereby stimulating further adoption. As consumer demand for authentic connections and impartial evaluations grows, the credibility and openness of social media influencers will assume greater significance. The incorporation of social commerce features is expected to enhance the assimilation procedure, facilitating the consumers exploration, evaluation, and procurement of novel technologies without any hindrance. The influence of user-generated content will persist as a significant factor,

and the provision of co-creation opportunities will enable consumers to exert a direct influence on the advancement of emerging technologies. As time progresses, the function of social media in the adoption of technology will persistently transform, influencing consumer conduct and fundamentally altering the approach towards the assimilation of novel technologies.

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Declarations

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